

Tutorial 00 — Part 5
Lattice Structures: SCC, BCC, FCC
2D and 3D plots + packing fractions

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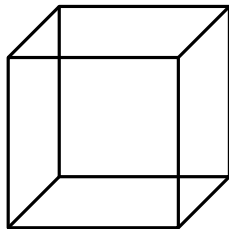
CH4114 — Molecular Simulation

14 August 2026

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Repository: `CH4114_Molecular_Simulation_Tutorials`

What is a crystal lattice?



unit cell, side a

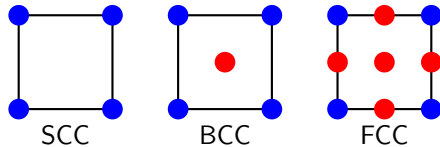
- A crystal is atoms in a **repeating pattern**.
- The smallest repeating unit is the **unit cell** — a cube of side a (the *lattice constant*).
- Three classic cubic lattices: **SCC**, **BCC**, **FCC**.

The three cubic lattices

Lattice	Atoms per cell	Radius r	Coordination
SCC	1 ($8 \text{ corners} \times \frac{1}{8}$)	$a/2$	6
BCC	2 ($\frac{1}{8} \text{ corners} + 1 \text{ center}$)	$a\sqrt{3}/4$	8
FCC	4 ($\frac{1}{8} \text{ corners} + 6 \text{ faces} \times \frac{1}{2}$)	$a\sqrt{2}/4$	12

- Neighboring atoms just **touch**:
- SCC along the *edge* ($2r = a$), BCC along the *body diagonal* ($4r = a\sqrt{3}$), FCC along the *face diagonal* ($4r = a\sqrt{2}$).

2D top views of the unit cells



- blue = corners, red = body/face centers.
- FCC seen from above: edge midpoints + the top-face center.

Packing fraction in 2D



- Packing fraction = area of atoms $\div$ total area.
- **Square lattice:** cell side $2r$, one circle per cell

$$\frac{\pi r^2}{(2r)^2} = \frac{\pi}{4} \approx 0.7854$$

- **Hexagonal lattice:** the densest 2D packing

$$\frac{\pi r^2}{2\sqrt{3} r^2} = \frac{\pi}{2\sqrt{3}} \approx 0.9069$$

3D unit cells with Matplotlib

```
def unit_cell_atoms(kind):
    corners = [(x, y, z)
               for x in (0, a) for y in (0, a)
               for z in (0, a)]
    if kind == "SCC":
        return np.array(corners)
    if kind == "BCC":
        return np.array(corners
                        + [(a/2, a/2, a/2)])
    if kind == "FCC":
        faces = [(0, a/2, a/2), (a, a/2, a/2),
                 (a/2, 0, a/2), (a/2, a, a/2),
                 (a/2, a/2, 0), (a/2, a/2, a)]
        return np.array(corners + faces)
```

- Scatter the atoms inside a cube.
- Draw the 12 cube edges so the cell is visible.
- One 3D subplot per lattice:
subplot(1,3,i,
projection="3d").

Drag to rotate

The 3D plot is interactive — rotate, zoom, inspect.

Packing fraction in 3D

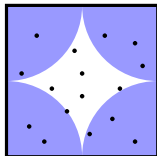
Lattice	Atoms	Radius	Packing fraction
SCC	1	$a/2$	$\pi/6 \approx 0.5236$
BCC	2	$a\sqrt{3}/4$	$\sqrt{3}\pi/8 \approx 0.6802$
FCC	4	$a\sqrt{2}/4$	$\pi/(3\sqrt{2}) \approx 0.7405$

Formula

$$\text{packing fraction} = \frac{\text{atoms per cell} \times \frac{4}{3}\pi r^3}{a^3}$$

- **FCC is the most efficient (74%)** — why copper, gold, aluminum crystallize in FCC.

Verify with Monte Carlo (random points)



- Throw N random points into the cube with `np.random`.
- Count how many fall **inside** an atom.
- Fraction inside $\approx$ packing fraction.

```
np.random.seed(42)
pts = np.random.rand(N, 3) * a
frac = inside_any(pts, atoms, r).mean()
print(frac)    # FCC -> ~0.7405
```

- The simulation agrees with the math!

Interactive 3D rotation

In a script

`plt.show()` opens a window — drag to rotate, scroll to zoom.

In a Jupyter notebook

```
%matplotlib widget      # interactive figures (needs: uv add --dev ipympl)
```

- **Drag** left button → rotate the lattice.
- **Right-drag** → zoom.
- **Scroll** → zoom in/out.
- Inspect the atom positions from every angle.

Scripts, notebooks, and check yourself

Run the companion code

```
uv run python examples/lattice_structures/lattice_2d.py
uv run python examples/lattice_structures/lattice_3d.py
uv run python examples/lattice_structures/packing_fraction.py
```

Check yourself

- 1 How many atoms does an FCC unit cell contain? Why 4, not 14?
- 2 Why is the BCC atom radius $a\sqrt{3}/4$ and not $a/2$?
- 3 Which lattice has the highest packing fraction?
- 4 What is the 2D packing fraction of a square lattice?
- 5 Why set `np.random.seed(...)` before the Monte Carlo check?